

Homework 1: The Sun

Solutions

Fall 2026

1 Radiation from the Corona

Question i) If the temperature in a region of the corona is 2 MK, what is the characteristic wavelength emitted by this region? What type of electromagnetic wave is this? ii) Repeat for a region of the photosphere at 5,778 K.

Answer The characteristic wavelength is the wavelength at the peak of the blackbody spectrum for a given temperature. Find the peak by taking the derivative of Planck's Law (which defines the blackbody spectrum) and finding where the derivative is equal to zero. This will give you a simple relationship which is known as Wien's Displacement Law:

$$\lambda = b/T$$

where

λ = wavelength [m]

T = object temperature [K]

b = a constant = 2.898×10^{-3} m K.

i) For the 2 MK corona, $\lambda = b/T \approx \boxed{1.45 \text{ nm}}$. This is an $\boxed{\text{X-ray}}$.

ii) For the 5,778 K case, $\lambda = b/T \approx \boxed{501 \text{ nm}}$. This is in the $\boxed{\text{Visible}}$ range.

Note that by using Wien's displacement law, derived from Planck's Law, we implicitly assumed that the gas is radiating with a blackbody spectrum. Is this a good assumption for the very hot corona? If not, what does it mean when we talk about the "temperature" of the corona??

2 Thermal escape

Question What is the thermal velocity of a proton in the outer corona where the temperature is 2 MK? (Hint: look up "thermal velocity" on wikipedia. Consider three dimensions.) Is this fast enough to escape the gravitational field of the sun? (Hint 2: look up "escape velocity".) Are coronal protons hot enough to escape from the sun's gravity?

Answer From the wiki page mentioned, we have three options for thermal speed, or a range that includes them all. Any are acceptable for this problem.

The *Most probable* speed is the mode of the thermal velocity distribution:

$$v_{th} = \sqrt{\frac{2k_B T}{m_p}} \approx \boxed{181.8 \text{ km/s}}$$

The *Root mean square* of the velocity distribution:

$$v_{th} = \sqrt{\frac{3k_B T}{m_p}} \approx \boxed{222.6 \text{ km/s}}$$

The *Mean* of the velocity distributions:

$$v_{th} = \sqrt{\frac{8k_B T}{m_p \pi}} \approx \boxed{205.1 \text{ km/s}}$$

These three alternate speeds give a *Range* of possible thermal velocities:

$$v_{th} = (1.6 \pm 0.2) \sqrt{\frac{k_B T}{m_p}} = \boxed{205.7 \pm 25.7 \text{ km/s}}$$

Hence, the choice of definition gives a value that is typically within 10–20% of the others. To compare to the escape velocity, you first need to calculate it! Use the formula:

$$v_e = \sqrt{\frac{2GM}{R}}$$

where

$$G = \text{universal gravitational constant} = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$$

$$M = \text{mass of the sun} = M_{\odot} = 1.99 \times 10^{30} \text{ kg}$$

$$R = \text{distance from the center of mass of the sun.}$$

You would want to use the radius corresponding to the location of the particle, but since this is not specified in the problem, you have to make some assumptions. You could use distances ranging from the surface of the sun, with $R_{\odot} = 695.7 \times 10^3 \text{ km}$, or any coronal radius, as long as you explain and justify your choice. Using $R_{\odot}$, the escape velocity becomes:

$$v_e = \sqrt{\frac{2 \times 6.67 \times 10^{-11} \times 1.99 \times 10^{30}}{695.7 \times 10^6}} = 6.18 \times 10^5 \text{ m/s} = \boxed{618 \text{ km/s}}$$

Hence, near the surface of the photosphere, **a proton at the average thermal velocity cannot escape the sun's gravitational field!**

However, as you move away from the sun, the escape velocity decreases; at $5R_{\odot}$, the escape velocity is reduced by a factor of $1/\sqrt{5}$ to 276 km/s. Now the thermal protons are much closer to the escape velocity; if they are moving away from the sun and do not encounter collisions, *some of them* may escape and become part of the solar wind. (Remember that the thermal velocity is characteristic of a distribution; some percentage of the particles in the tail of the distribution will be faster than this escape velocity.)

At 1 AU, the escape velocity from the sun is only 42.1 km/s. Note that the solar wind is moving much faster than this; this implies that by the time it reaches Earth, the solar wind is not significantly under the gravitational influence of the sun. (This is important: as the solar wind reaches further out into the solar system, to Jupiter's orbit and beyond, the velocity does not change much!)

3 Solar Wind

Question i) The quiescent solar wind density at Earth is about 5 ions/cm³, moving at a velocity of 400 km/s. If most of these ions are protons, what is the total mass ejected from the sun per second? ii) Given these parameters, what is the solar wind particle flux (ions/cm²/sec) at the orbit of Jupiter?

Answer i) We can assume that no solar wind mass is lost between when it is ejected from the Sun to when it reaches 1 AU. Since we know the density and speed at 1 AU, we want to find the total solar wind mass that flows through the sphere created at 1 AU in one second. First, let's convert the density to ions m⁻³:

$$5 \text{ ions cm}^{-3} \times \frac{100^3 \text{ cm}^3}{1 \text{ m}^3} = 5 \times 10^6 \text{ ions m}^{-3}$$

Next, we convert from number density (ions m⁻³) to mass density (kg m⁻³) with the assumption that the majority of ions are protons ($m_p = 1.67 \times 10^{-27} \text{ kg}$):

$$5 \times 10^6 \text{ protons m}^{-3} \times 1.67 \times 10^{-27} \text{ kg/proton} = 8.35 \times 10^{-21} \text{ kg m}^{-3}$$

To see how much mass is flowing through a unit of area at any given second, we multiply by the speed:

$$8.35 \times 10^{-21} \text{ kg m}^{-3} \times 400,000 \text{ m s}^{-1} = 3.34 \times 10^{-15} \text{ kg s}^{-1} \text{ m}^{-2}$$

And finally, we need to look at the entire area the wind is flowing through—the sphere the solar wind makes at 1 AU:

$$3.34 \times 10^{-15} \text{ kg s}^{-1} \text{ m}^{-2} \times 4\pi(1.496 \times 10^{11} \text{ m})^2 = \boxed{9.39 \times 10^8 \text{ kg s}^{-1}}$$

This entire process can be completed in one step as with the equation $\rho v(4\pi r^2)$. If you do not include the 4π , your answer will be off by an order of magnitude!

ii) First, get the flux at Earth:

$$\begin{aligned} 5 \times 10^6 \text{ ions m}^{-3} \times 400,000 \text{ m s}^{-1} &= 2 \times 10^{12} \text{ ions m}^{-2} \text{ s}^{-1} \\ &= 2 \times 10^8 \text{ ions cm}^{-2} \text{ s}^{-1} \end{aligned}$$

You might have found this in part (i) if you followed the steps in a different order. Then, get the flux at Jupiter with the inverse square law:

$$\begin{aligned} \frac{\text{Flux}_\oplus}{\text{Flux}_J} &= \frac{r_J^2}{r_\oplus^2} \\ \frac{2 \times 10^8 \text{ ions cm}^{-2} \text{ s}^{-1}}{\text{Flux}_J} &= \frac{(5.2 \text{ AU})^2}{(1 \text{ AU})^2} \\ \text{Flux}_J &= \boxed{7.4 \times 10^6 \text{ ions cm}^{-2} \text{ s}^{-1}} \end{aligned}$$

Note that you could instead apply the inverse square law first, then convert to the particle flux, and arrive at the same answer.

Check and see if the answer makes sense. Should the flux at Jupiter be smaller than Earth? Yes! The flux decreases with $1/r^2$ as the solar wind particles radiate from the Sun. The same amount of mass reaches 5.2 AU, but it is spread over a much larger sphere.

4 Relativistic Protons

Question Assuming that most solar protons are accelerated in the low corona ($\sim 2R_S$), create a plot that shows distance traveled in AU vs time for 1, 10, 100, and 500 MeV protons. How long in hours does it take for each of these energies to reach Earth? (refer to USW Section 2.1.4).

Answer Referencing USW Section 2.1.4, you can start from Equation 2-13 and solve for the velocity:

$$KE = \left(\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} - 1 \right) m_0 c^2$$

$$v = c \sqrt{1 - \frac{1}{\left(\frac{KE}{m_0 c^2} + 1 \right)^2}}$$

Using $c = 3 \times 10^8$ m/s, $m_p c^2 \approx 940$ MeV as the rest energy of a proton, and plugging in each energy (can keep in MeV units, as they will cancel), we find:

$$v_{1 \text{ MeV}} = 1.38 \times 10^7 \text{ m/s}$$

$$v_{10 \text{ MeV}} = 4.34 \times 10^7 \text{ m/s}$$

$$v_{100 \text{ MeV}} = 1.28 \times 10^8 \text{ m/s}$$

$$v_{500 \text{ MeV}} = 2.27 \times 10^8 \text{ m/s}$$

Taking the distance of 1 AU in meters ($\sim 1.496 \times 10^{11}$ m) (I recommend you to remember: 1 Au $\sim 215R_\odot$, $1 R_\odot \sim 109 R_E$, thus, $1 \text{ Au} \simeq 215 \times 109 R_E$) and dividing it by the proton velocity should give us the time it takes the proton to travel to Earth (assuming they travel directly to Earth):

$$t_{1 \text{ MeV}} = \boxed{3.0 \text{ hrs}}$$

$$t_{10 \text{ MeV}} = \boxed{0.96 \text{ hrs}}$$

$$t_{100 \text{ MeV}} = \boxed{0.32 \text{ hrs}}$$

$$t_{500 \text{ MeV}} = \boxed{0.18 \text{ hrs}}$$

Your plot should look like Figure 1. You may also use 1 AU - $2R_\odot$ as your distance, which will give you results quite close to those found using 1 AU above:

$$t_{1 \text{ MeV}} = 2.98 \text{ hrs}$$

$$t_{10 \text{ MeV}} = 0.95 \text{ hrs}$$

$$t_{100 \text{ MeV}} = 0.32 \text{ hrs}$$

$$t_{500 \text{ MeV}} = 0.18 \text{ hrs}$$

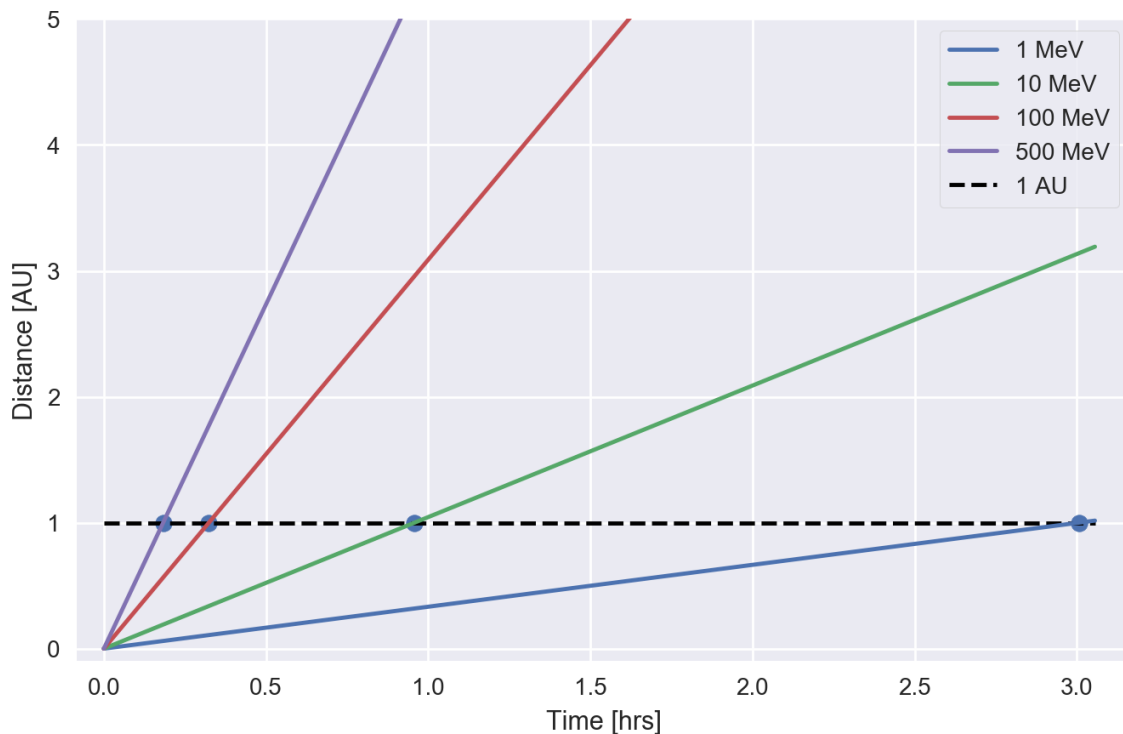


Figure 1: Time for protons of different energies to reach distances from the sun.

5 Heliopause

Question The Heliopause occurs where the pressure of the interstellar medium balances the pressure of the solar wind. If the interstellar pressure is $\sim 1.3 \times 10^{-13}$ Pa, calculate the distance from the sun to the heliopause in AU. Again assume the solar wind density at 1AU is 5 protons/cm³.

Answer By definition, the heliopause is located where the ram pressure of the solar wind (SW) is equal to the external pressure of the interstellar medium (ISM):

$$\begin{aligned}
 P_{ISM} &= P_{SW} \\
 P_{ISM} &= \rho v^2 \\
 P_{ISM} &= m_p n v^2
 \end{aligned}$$

Where does the distance to the heliopause come into this equation? In the number density! We're back to the inverse square law again to relate density at Earth to density at the heliopause:

$$\frac{n_{\oplus}}{n_{hp}} = \frac{r_{hp}^2}{r_{\oplus}^2} \rightarrow n_{hp} = n_{\oplus} \frac{r_{\oplus}^2}{r_{hp}^2}$$

So let's plug it in and solve for r_{hp} ! (If we keep everything in SI units, but put $r_{\oplus}$ in AU, we will

get r_{hp} in AU.)

$$P_{ISM} = m_p n_{\oplus} \frac{r_{\oplus}^2}{r_{hp}^2} v^2 \rightarrow r_{hp} = \sqrt{\frac{m_p n_{\oplus} r_{\oplus}^2 v^2}{P_{ISM}}}$$

$$r_{hp} = \sqrt{\frac{(1.67 \times 10^{-27} \text{ kg})(5 \times 10^6 \text{ m}^{-3})(1 \text{ AU})^2(400,000 \text{ m s}^{-1})^2}{1.3 \times 10^{-13} \text{ Pa}}}$$

$$\approx \boxed{101.4 \text{ AU}}$$

6 Spacecraft Temperature

Question Assume that aluminum has a solar absorptivity of 0.15 and an emissivity of 0.05. i) Imagine a spherical probe orbiting the Earth; what is its equilibrium temperature when it is in sunlight? ii) What about the same probe orbiting Jupiter? iii) and iv) Repeat, but now assume the spacecraft is coated with GSFC White Paint NS-74, which has a solar absorptivity of 0.17 and emissivity of 0.92. What do you think this paint is designed for?

Answer Here is the equilibrium temperature of a sphere in space exposed to the sun.

$$T = \left(\frac{\alpha}{\epsilon}\right)^{1/4} \left(\frac{SA_n}{\sigma A_{tot}}\right)^{1/4} = \left(\frac{\alpha}{\epsilon}\right)^{1/4} \left(\frac{S\pi r_{SC}^2}{\sigma 4\pi r_{SC}^2}\right)^{1/4} = \left(\frac{\alpha}{\epsilon}\right)^{1/4} \left(\frac{S}{4\sigma}\right)^{1/4}$$

For part (i), $\alpha = 0.15$, $\epsilon = 0.05$, $S = 1361 \text{ W/m}^2$ (solar flux at 1 AU), and $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2/\text{K}^4$ (Stefan-Boltzmann constant).

$$T = \left(\frac{0.15}{0.05}\right)^{1/4} \left(\frac{1361 \text{ W m}^{-2}}{4 \times 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}}\right)^{1/4} \approx \boxed{367 \text{ K}}$$

For part (ii) only the value for S changes. We use the inverse square law to get S at Jupiter and then solve for the temperature again:

$$S = \frac{1361 \text{ W/m}^2}{5.2^2} \approx 50.3 \text{ W/m}^2$$

$$T = \left(\frac{0.15}{0.05}\right)^{1/4} \left(\frac{50.3 \text{ W m}^{-2}}{4 \times 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}}\right)^{1/4} \approx \boxed{161 \text{ K}}$$

For parts (iii) and (iv), repeat with new α and ϵ :

$$T = \left(\frac{0.17}{0.92}\right)^{1/4} \left(\frac{1361 \text{ W m}^{-2}}{4 \times 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}}\right)^{1/4} \approx \boxed{183 \text{ K}}$$

$$T = \left(\frac{0.17}{0.92}\right)^{1/4} \left(\frac{50.5 \text{ W m}^{-2}}{4 \times 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}}\right)^{1/4} \approx \boxed{80.1 \text{ K}}$$

The coating significantly reduced the equilibrium temperature! It is likely used as a passive cooling method of thermal control. For almost all CubeSats, e.g., CSSWE and CIRBE, we used passive thermal control. I have not heard any CubeSat using active thermal control. Some bigger mission, such as Parker Solar Probe, is using active thermal control.

7 Solar Outputs

Question Use Figure 22 (in Module1 notes) to answer the following questions:

- (a) What is the minimum radiative energy flux of an M-Class flare?
- (b) Consider the flare at the beginning of March 07, 2012. Assume the peak energy flux on this plot is $5 \times 10^{-4} \text{ W/m}^2$ (it's a bit hard to see because of the text). Determine the corresponding energy flux at the emission source location of 1.5 solar radii.
- (c) The flares with the sharp rise and multi-hour decay (long duration events–LDEs) are the one most **likely** to have an associated Coronal Mass Ejection (CME). About how many CMEs were likely launched into the heliosphere according to this chart?
- (d) Comment on the relationship between the X-ray flares (top panel) and the flux of solar protons (second panel).
- (e) Comment on the relationship between cosmic rays (the third panel) and the two panels above it. (We'll ignore the magnetic field signatures until the next module.)

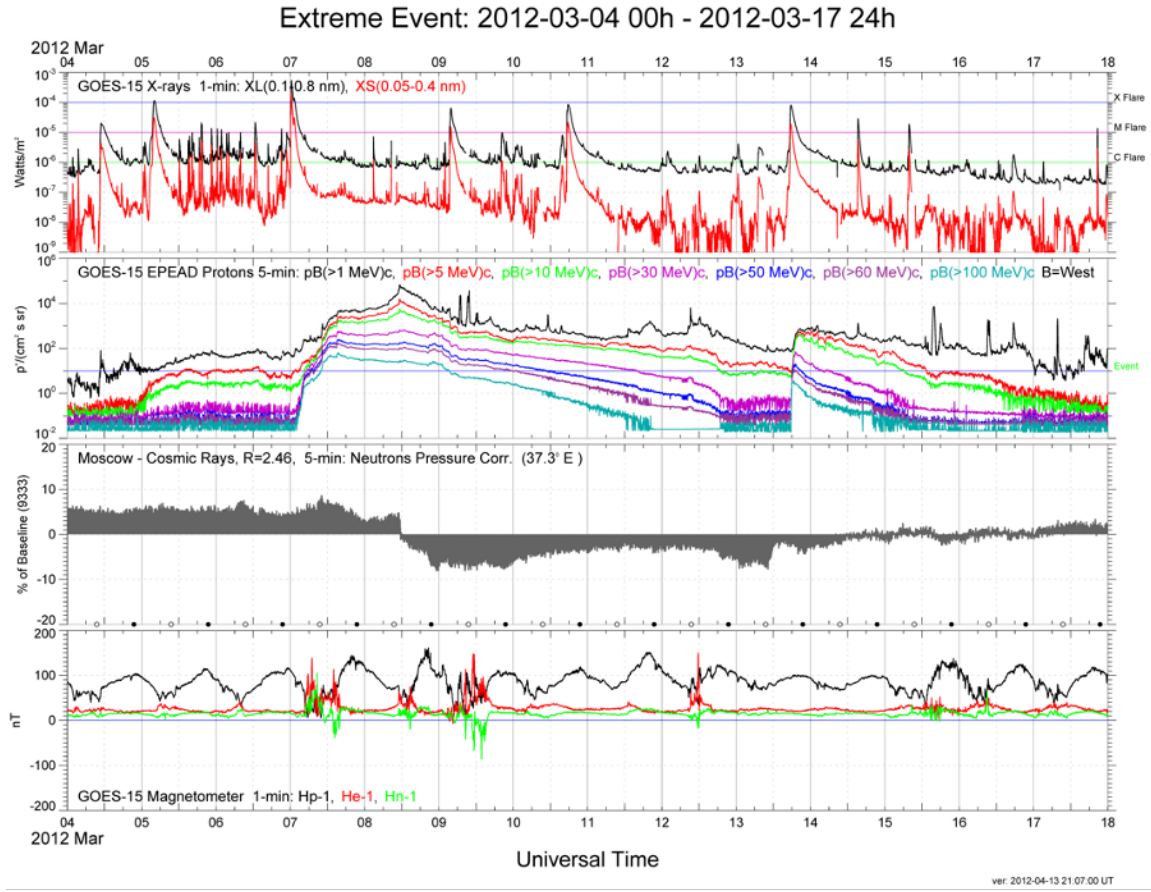


Figure 2: Outputs from the sun in March 2012. From top to bottom: i) Radiative energy flux measured at 1 AU in two wavelength bands; ii) energetic protons measured at 1 AU in a range of energy bands; iii) cosmic ray flux variation measured on the ground near Moscow; and iv) three-axis magnetic field intensity measured by GOES at 1 AU.

Answer

- (a) The minimum radiative energy flux of an M-Class flare is 10^{-5} W/m^2 .
- (b) You can use the relationship for flux conversion through a sphere (USW Equation 2-19), which easily reduces to the inverse square law:

$$\begin{aligned}
 4\pi r_i^2 I_i &= 4\pi r_f^2 I_f \\
 \frac{I_i}{I_f} &= \frac{r_f^2}{r_i^2} \\
 \frac{5 \times 10^{-4} \text{ W/m}^2}{I_f} &= \frac{(1.5R_s)^2}{(215R_s)^2} \\
 \frac{5 \times 10^{-4} (215)^2}{(1.5)^2} &= 10.272 \text{ W/m}^2
 \end{aligned}$$

- (c) There are about 6 CMEs likely to have been launched into the heliosphere during the two-week period shown.
- (d) The sharp increases in proton flux are solar energetic particle events (SEPs), which occur when solar flares or coronal mass ejections accelerate coronal and interplanetary ions and electrons to relativistic energies. There is a sharp increase in solar protons following most of the major flares, but not all. The SEP events also vary in terms of which energies are enhanced after each flare. And, while some of the flares are associated with brief increases in proton flux (e.g., the event on March 4), there are clearly two prominent SEP events (March 7 and March 13) where all energy channels are dramatically enhanced, and the flux is raised for many days.
- (e) The third panel measures the cosmic ray flux reaching the Earth's surface at a detector site near Moscow. Your intuition might tell you that the sun's activity should not affect the cosmic ray flux. Indeed, there are not sharp, immediate responses in the cosmic ray baseline percentage corresponding to the flares and SEPs like in part (d). However, we can see that the baseline percentage of cosmic rays changes suddenly by $\sim 10\%$ after the proton fluxes reach close to their peak on March 8th. You'll also notice that the GOES Magnetometer (bottom panel) is also showing some activity at this time. Most likely there was a CME at this time which perturbed the Earth's magnetosphere, which has some hindrance over the cosmic rays reaching the Earth's surface at different latitudes. We'll learn more about this in the next modules!

8 Parker Solar Probe

Question Parker Solar Probe has five “investigations”: [FIELDS](#), [ISIS](#), [WISPR](#), [SWEAP](#), and [HeliOSPP](#). Each of these investigations includes multiple instruments. Do some research and pick your favorite instrument. In a couple of paragraphs, explain:

- [What does this instrument do?](#)
- [What is the instrument SWAP \(size, weight, and power\)?](#)
- [In basic terms, how does it work?](#)
- [How does it contribute to the science goals of its parent investigation?](#)

Answer Varies by selected instrument. Slack us if you have questions about your responses!
Here's a basic outline of the details you want to understand about an instrument:

- [What does this instrument do?](#) What does it measure: electric fields, magnetic fields, particles, photons? What is the energy or frequency range to which it is sensitive? How does that sensitivity vary? What is the principle by which it makes its measurements? What is its FOV, or directionality? What is the time resolution, spatial resolution, energy and/or frequency resolution of the final data products?
- [What is the instrument SWAP \(size, weight, and power\)?](#) Is this an instrument that requires a big spacecraft (that can carry a lot of mass, and provide a lot of power), or could it be accommodated on a future CubeSat or SmallSat mission?
- [In basic terms, how does it work?](#) See above: what is the physical principle that produces the measurement we want?

- **How does it contribute to the science goals of its parent investigation?** How do the instrument and its particular observations tie into the mission goals? For example, do we need these particles at these energies, together with data from another instrument, to fully answer mission science question #X?